

8.2 Polyphase Implementation of Decimation and Interpolation Filters Using MATLAB

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clc;

clear;

close all;

%% Input Signal

Fs = 1000; % Sampling Frequency

t = 0:1/Fs:1; % Time Vector

% Test Signal

x = sin(2*pi*50*t) + 0.5*sin(2*pi*120*t);

%% FIR Low-Pass Filter Design

N = 20; % Filter Order

Fc = 0.4; % Normalized Cutoff Frequency

h = fir1(N,Fc,hamming(N+1));

%% Decimation Factor

M = 2;

% Polyphase Decimation

y_dec = upfirdn(x,h,1,M);

%% Interpolation Factor

L = 2;

% Polyphase Interpolation

y_int = upfirdn(y_dec,h,L,1);

%% Adjust Length for Comparison

if length(y_int) > length(x)

    y_int = y_int(1:length(x));

elseif length(y_int) < length(x)

    x = x(1:length(y_int));

end
```

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%% Time Axes
t1 = (0:length(x)-1)/Fs;
t2 = (0:length(y_dec)-1)/(Fs/M);
t3 = (0:length(y_int)-1)/Fs;

%% BEST GRAPH OUTPUT
figure('Name','Polyphase Decimation and Interpolation','NumberTitle','off');

% Original Signal
subplot(4,1,1)
plot(t1,x,'b','LineWidth',1.5)
title('Original Input Signal')
xlabel('Time (s)')

ylabel('Amplitude')
grid on

% Decimated Signal
subplot(4,1,2)
stem(t2,y_dec,'r','filled')
title('Decimated Signal (M = 2)')
xlabel('Time (s)')
ylabel('Amplitude')
grid on

% Interpolated Signal
subplot(4,1,3)
plot(t3,y_int,'k','LineWidth',1.5)
title('Interpolated/Reconstructed Signal (L = 2)')
xlabel('Time (s)')
ylabel('Amplitude')
grid on

% Comparison

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subplot(4,1,4)
plot(t1,x,'b','LineWidth',1.5)
hold on
plot(t3,y_int,'r--','LineWidth',1.5)
title('Comparison of Original and Reconstructed Signals')
xlabel('Time (s)')
ylabel('Amplitude')
legend('Original Signal','Reconstructed Signal')
grid on

%% Display Information
disp('-----');
disp('POLYPHASE DECIMATION & INTERPOLATION');

disp('-----');
fprintf('Original Signal Length = %d\n',length(x));
fprintf('Decimated Signal Length = %d\n',length(y_dec));
fprintf('Interpolated Signal Length = %d\n',length(y_int));

%% Separate Figure for Better Visualization
figure;
plot(t1,x,'b','LineWidth',2)
hold on
plot(t3,y_int,'r--','LineWidth',2)
title('Original vs Reconstructed Signal')
xlabel('Time (s)')
ylabel('Amplitude')
legend('Original Signal','Reconstructed Signal')
grid on

```

Polyphase Decimation and Interpolation

FIGURE

FORMAT

TOOLS

LAYOUT



